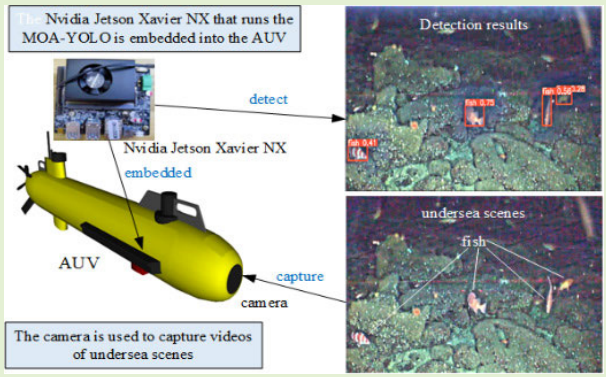
MOA-YOLO: An Accurate, Real-Time, and Lightweight YOLOv10-Based Algorithm for Deep-Sea Fish Detection

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***Abstract*—The use of autonomous underwater vehicles (AUVs) is an efficient mean to detect deep-sea fishes due to their great ability to complete undersea mission autonomously. However, research on deep-sea fishes suffers from dynamic and small objects in degraded environments, making it difficult to recognize deep sea. In this article, an accurate, real-time, and lightweight deep-sea fish detection algorithm is proposed based on YOLOv10, named MOA-YOLO. First, an underwater image enhancement algorithm based on adaptive standardization and normalization networks is adopted to correct the color and contrast of deep-sea images. Then, the multihead latent attention (MLA) mechanism is added to the front of each detection head of MOA-YOLO to extract more object features, enhancing the discrimination between objects and surroundings. Next, an optimized normalized Wasserstein distance (ONWD) loss function is introduced to replace the complete intersection over union (CIOU) loss function, improving the ability to detect small objects. Finally, CBS and C2f modules in the backbone and neck are replaced with alterable kernel convolution (AKConv) and AKC2f modules, respectively, balancing the detection speed and accuracy. The comparative experiments are operated on both the desktop computer and Nvidia Jetson Xavier NX. Compared with mainstream algorithms, the precision, recall, mean average precision (mAP_{50}), $mAP_{50:95}$, and frames per second (FPS) of MOA-YOLO are improved by at least 3.1%, 13.7%, 8.0%, 6.9%, and 8.8%, respectively. The experimental results prove that MOA-YOLO achieves excellent performance on deep-sea fish detection with the AUV. The code is available at https://github.com/hu167/code**



***Index Terms*— Alterable kernel convolution (AKConv), deep-sea fish detection, loss function, multihead latent attention (MLA), underwater image enhancement.**

I. Introduction

THE deep sea contains various species of deep-sea fishes such as cod, tuna, and salmon, which have rich research values in marine fishery and food [1]. However, the degraded underwater environment makes it difficult to catch sight of deep-sea fishes. Moreover, deep-sea fishes are small and dynamic objects that are easily confused with their surroundings. Thus, deep-sea fishes exhibit quick and erratic movements to cause frequent negatives in detection due to their small sizes coupled with low visibility [2], [3]. Artificial detection is unsuitable for deep-sea fish detection due to the degraded environment and characteristics of deep-sea fishes. Autonomous underwater vehicles (AUVs) that have been widely applied to oceanographic surveys can be a useful tool for deep-sea detection. However, deep-sea fish detection requires accurate and real-time performance due to the dynamic and small objects of fishes. Therefore, it is essential to design an effective real-time detection method to detect deep-sea fishes.

With the constant development of deep learning technology, deep neural networks have been widely applied to underwater target detection. At the same time, an increasing number of deep learning algorithms have been proposed to detect marine fishes. However, the AUV that contains limited computing resources can only use low-power processors such as the Nvidia Jetson Xavier NX, which can only operate limited parameters [4], [5]. Therefore, it is necessary to propose an accurate, lightweight, and real-time detection algorithm that can be deployed on the AUV.

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You only look once (YOLO) is a single-stage algorithm, which produces a high detection speed. YOLO series have evolved from the original generation of YOLOv1 to the latest generation of YOLOv13. Since YOLOv11, C2f modules are simplified, which may cause poor performance of small target detection. In contrast, YOLOv10 that contains full C2f modules can recognize small targets rapidly. Thus, YOLOv10 is suitable for the detection of deep-sea fishes. Nevertheless, due to the limited computational resources, embedded image processors such as Nvidia Jetson Xavier NX cannot run the original YOLOv10 in real time. Therefore, the MOA-YOLO algorithm combined with an underwater image enhancement algorithm is proposed to improve the detection speed and accuracy of deep-sea fishes in this article. To correct the color and contrast of deep-sea images, an underwater image enhancement algorithm based on adaptive standardization and normalization networks (ASNets) is adopted. Compared with the original YOLOv10, the multihead latent attention (MLA) mechanism is added to the front of each detecting head to distinguish objects from the surroundings. Additionally, an optimized normalized Wasserstein distance (ONWD) loss function is proposed to replace the original complete intersection over union (CIOU) loss function to detect small objects. Furthermore, alterable kernel convolution (AKConv) and AKC2f modules are introduced into the backbone and neck of YOLOv10 to improve the detection speed while maintaining the detection accuracy. The procedure of the deep-sea fish detection system with the AUV is organized as follows. First, the camera on the AUV captures videos of undersea scenes and transmits them to the Nvidia Jetson Xavier NX embedded in the AUV. Then, the MOA-YOLO deployed on the Nvidia Jetson Xavier NX detects deep-sea fishes from the captured images in real time.

The main improvements of this study are as follows.

1) The underwater image enhancement algorithm based on ASNet is adopted to remove color basis and improve the contrast of underwater images to obtain enhanced images, which can be used to improve the detection performance of deep-sea fishes.
2) In the neck, the MLA is added to the front of each detecting head to extract object features in multiple attention heads with group query attention, allowing for extracting more effective features of deep-sea fishes from the surroundings.
3) In the detecting head, an ONWD loss function is proposed to replace the CIOU loss function to improve the detection accuracy of small targets. The ONWD utilizes normalized Wasserstein distance (NWD) to weaken the impact on the intersection over union (IOU) for small targets while combining IOU and CIOU loss functions to calculate deviations of bounding features, positions, and shapes between bounding boxes, promoting to detect small targets.
4) In the backbone and neck, AKConv and AKC2f modules are adopted to replace CBS and C2f modules, respectively. Both the AKConv and AKC2f modules use flexible parameter numbers, sampling sizes, and sampling shapes to extract image features, making it efficient to reduce parameters and improve feature extraction ability. Thus, the balance between detection speed and accuracy can be achieved.

II. Related Works

Recently, more and more researchers have used deep learning networks to detect small and dynamic underwater targets like fishes. Vision-based underwater target detection methods are mainly based on convolutional neural network (CNN) algorithms, which can be classified into two-stage methods and one-stage methods.

A. Two-Stage Methods

Two-stage methods select regions that may contain objects and then detect objects on these regions. The region-based CNN (R-CNN) [6] is a typical two-stage method that achieves an excellent framework for target detection. Fan [7] improved the Faster R-CNN [8] by adding more anchors and channels into the region proposal network to detect fishes. The results show that the detection accuracy is increased by over 6.23% compared with the original Faster R-CNN. Lu et al. [9] proposed an improved Faster R-CNN algorithm for underwater object detection by replacing the deeper neural network ResNet-101 with the VGG-16 network, producing a precision of 63.03%. Zeng et al. [10] proposed a Faster R-CNN-AON algorithm by introducing an adversarial occlusion network to enhance the detection effect in various underwater environments. Yu and Shuiyuan [11] improved the Cascade R-CNN [12] to detect underwater targets by adjusting hyperparameters, significantly improving the detection accuracy of small targets. However, the two-stage methods will result in a long computational time.

B. One-Stage Methods

One-stage methods can effectively speed up the detection process due to their relatively lightweight models. Qiang et al. [13] improved the single-shot multibox detector (SSD) [14] by replacing the visual geometry group network (VGGNet) with residual network (ResNet) to improve the detection speed in complex water environments. However, the recall of SDD is not satisfied owing to the lack of low-level feature convolution layers. On the contrary, YOLO is a typical one-stage detection technology, which reduces computational time and keeps a high detection precision and recall. Pei et al. [15] proposed an improved YOLOv3 [16] algorithm for real-time object detection in an engineering fishway. Mohamed et al. [17] integrated image enhancement into YOLOv3 to improve the fish detection accuracy in turbid water environments. This method is operated on a real dataset and obtains a precision of 0.897 and 240 frames/s. Compared with the existing detection methods, each evaluation metric of the proposed method is increased by 10%–20% on average. Zhao et al. [18] improved the YOLOv4 [19] algorithm by replacing the backbone network with a lightweight network MobileNetV3 to detect underwater objects. The experimental results show that the algorithm achieves high detection accuracy. Li et al. [20] integrated the deformable convolution module (DCM)

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and adaptive threshold module (ATM) into YOLOv5 [21] to detect farmed fishes. This algorithm achieves a high detection precision owing to the heavyweight network. Li et al. [22] introduced the coordinate attention, multiscale detection layer, and loss function efficient IOU (EIOU) into the YOLOv5 model for fish detection. Yu et al. [23] added a cross-attention convolution and a squeeze–excitation module into YOLOv7 [24] to detect underwater organisms. Nevertheless, the algorithm is unsuitable for deploying on portable devices due to its large number of parameters. Liu et al. [25] proposed a fish detection method based on a lightweight YOLOv8 model to improve detection speed. Li et al. [26] introduced focal loss and adaptively spatial feature fusion (ASFF) into YOLOvX-nano to recognize underwater fishes, achieving an average precision of 87.7% and a parameter number of only 2.27 M. Wang et al. [27] improved the YOLOv10 [28] algorithm by adding a NVD loss function, a C2f-D-LKA layer and a DySample upsampling structure for fish skin health detection. The results show that this algorithm displays accurate and real-time performance of fish detection. Jalal et al. [29] combined YOLOv11 [30] with a custom-designed lightweight motion-based segmentation model to detect dynamic fishes while suppressing background interference. This improved YOLOv11 algorithm achieves the $F1$ value of 90.0% and 83.7% on the OzFish and Fish4Knowledge LifeCLEF 2015 datasets, respectively. However, YOLOv11 is not suitable for small object detection, especially in degraded backgrounds such as deep-sea environments.

III. Methods

A. YOLOv10 Algorithm

YOLOv10 is the tenth generation of YOLO series. The structure of YOLOv10 is shown in Fig. 1. YOLOv10 is divided into five types: YOLOv10n, YOLOv10s, YOLOv10m, YOLOv10l, and YOLOv10x. Compared with the other YOLOv10 models, YOLOv10s has a relatively smaller model size. YOLOv10s is made up of four parts: the input, backbone, neck, and head, defined as follows.

1) *Input:* The input includes the methods of dataset enhancement, image size adjustment, and adaptive anchor box calculation. The original image is converted into a three-channel image with the size of 640 × 640 by the three methods.
2) *Backbone:* The backbone comprises the CBS, SCDown, C2f, C2fCIB, spatial pyramid pooling-fast (SPPF), and partly self-attention mechanism (PSA) to extract important features from images. The C2f and C2fCIB components are the replacement of the CSPNet [31] component to achieve more accurate detection. The SCDown is the improved version of CBS to reserve more image information.
3) *Neck:* The neck located between the backbone and head adopts a feature pyramid network (FPN) [32] and a path aggregation network (PAN) [33]. The FPN with the method of upsampling and the PAN with a bottom–up pyramid are collectively used to extract and fuse the shallow and deep feature information in the three feature layers, forming feature maps with enhanced features repeatedly.
4) *Head:* The head is used to obtain the position of the prediction box in the image through the scale-sensitive IOU (CIOU) [34] loss function.

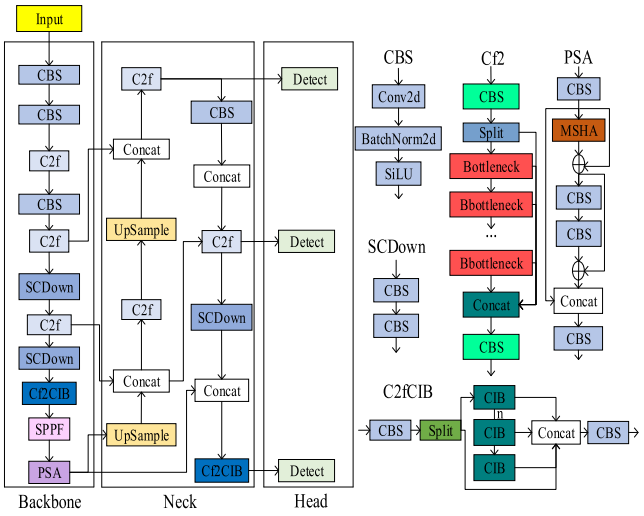


Fig. 1. Flowchart of YOLOv10.

The neck of the original YOLOv10 lacks attention mechanisms, leading to poor feature extraction ability. The loss function CIOU in YOLOv10 limits the ability to detect small objects due to the obvious impact on the IOU for small targets. Additionally, the CBS and C2f modules are not lightweight enough, resulting in a low detection speed. Thus, it is essential to further improve the original YOLOv10 to satisfy deep-sea fish detection.

B. Proposed Algorithm

This article proposes an improved YOLOv10 algorithm called MOA-YOLO for accurate, real-time, and lightweight deep-sea fish detection by AUVs in degraded sea environments. First, the underwater image enhancement algorithm based on ASNet is used to restore the color and contrast of the deep-sea images. Then, the MLA is introduced into the front of each detecting head to extract more useful image features, making deep-sea fishes easy to distinguish from the surroundings. Next, the CIOU loss function is replaced with an ONWD loss function, which effectively diminishes the impact of small targets on the IOU to detect small targets more effectively. Finally, CBS and C2f modules in the backbone and neck are replaced with AKConv and AKC2f modules, respectively, achieving a great balance between detection speed and accuracy. The flowchart of the proposed algorithm is shown in Fig. 2.

1) Adaptive Standardization and Normalization Networks: The ASNet [35] comprises adaptive standardization networks (ASNs) and adaptive normalization networks (ANNs) to correct the color and contrast of underwater images. The framework of ASNet is shown in Fig. 3.

The ASN is made up of a convolution layer, batch normalization (BN) [36], and a sigmoid activation function. The

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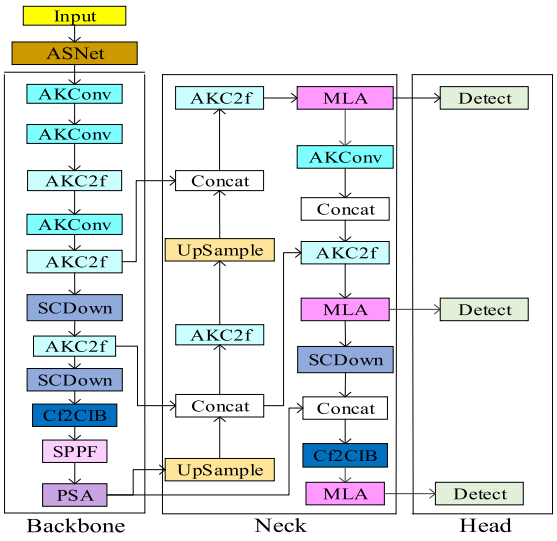


Fig. 2. Flowchart of MOA-YOLO.



Fig. 3. Framework of ASNet.

output $\mathbf{U}$ is calculated by the input image feature $\mathbf{F}$

$$\mathbf{U} = \mathbf{v}_b * \mathbf{F} \tag{1}$$

$$\mathbf{F} = [\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_a], \quad a \in \{1, 2, \ldots, A\} \tag{2}$$

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_b]$$

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_b], \quad b \in \{1, 2, \ldots, B\} \tag{3}$$

where $*$ represents a convolution operation and $\mathbf{v}_b$ is the dimension of the ath convolution filter $\mathbf{U}$. The first input of the ASN is an underwater image with $A = 3$, and the input dimension of the remaining ASN is B.

Well-distributed color channel histogram can be achieved by the standardization operation, which adopts the mean and standard deviations to scale features. BN is a linearized version of feature standardization that produces higher learning rates and a less meticulous initialization. BN generates the normalized feature map $\mathbf{n}_b$ of each $\mathbf{u}_b$. $\mathbf{n}_b(x, y)$ and $\mathbf{u}_b(x, y)$ represent the feature values of $\mathbf{n}_b$ and $\mathbf{u}_b$ at the spatial location of (x, y), respectively. Furthermore, $\mathbf{n}_b(x, y)$ is calculated by the mean of $\mathbf{u}_b$ ($\mu(\mathbf{u}_b)$) and variance of $\mathbf{u}_b(\text{var}(\mathbf{u}_b))$

$$\mathbf{n}_b(x, y) = \alpha_b^2 \beta_b \frac{\mathbf{u}_b(x, y) - \mu(\mathbf{u}_b)}{\sqrt{\text{var}(\mathbf{u}_b) + \varepsilon}} + \alpha_b \beta_b^2 \tag{4}$$

where α_b and β_b are adjustable parameters and ε is a small value to circumvent division by zero. The values of α_b and β_b are optimized via the gradient descent to minimize $\mathbf{n}_b(x, y)$. The gradient descent iteratively adjusts parameters to converge toward local minima, minimizing values of objective functions. Thus, α_b and β_b are calculated by the following formulas:

$$\frac{\partial \mathbf{n}_b(x, y)}{\partial \alpha_b} = 0, \quad \frac{\partial \mathbf{n}_b(x, y)}{\partial \beta_b} = 0. \tag{5}$$

Finally, the output of the ASN is

$$\mathbf{S} = f_s(\mathbf{F}) = \sigma(\mathbf{C}) \tag{6}$$

where f_s is the function of the ASB; C is the standardized feature with 128 channels; and $\mathbf{S}$ and $\mathbf{C}$ are $[\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_B]$ and $[\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_B]$, respectively. The sigmoid activation function σ is used to constrain the range of the output between 0 and 1.

The ANN changes the range of feature values by calculating the maximum and minimum values with two squeeze excitation (SE) [37] blocks: the maximum pooling (MaSE) and minimum pooling (MiSE), respectively. The feature $\mathbf{F}$ is normalized as

$$\mathbf{f}_n(i, j) = \frac{\mathbf{f}(i, j) - \min_p(\mathbf{f}(i, j))}{\max_p(\mathbf{f}(i, j)) - \min_p(\mathbf{f}(i, j))} \tag{7}$$

where $\mathbf{f}_n$ denotes the normalized feature and $\max_p$ and $\min_p$ represent the percentile maximum and percentile minimum, respectively.

The SE block contains squeeze and excitation. The squeeze is used to squeeze global spatial information with global average pooling. The excitation, which is made up of a fully connected layer, a ReLU function, and a sigmoid function, is used to fully capture channel-wise dependencies. The SE adopts global average pooling based on the input feature $\mathbf{f}_c$ extracted by a convolution layer

$$\mathbf{z}_c = \mathbf{F}_{\text{sq}}(\mathbf{f}_c) \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbf{f}_c(i, j) \tag{8}$$

where $\mathbf{z}_c$ is the squeezed global spatial information, $H \times W$ represents the size of the feature, and $\mathbf{F}_{\text{sq}}$ is the squeeze function using the global average operation.

The gating mechanism G is used to aggregate the image features during the squeeze operation

$$\boldsymbol{\omega} = \mathbf{F}_{\text{ex}}(\mathbf{z}, \mathbf{W}) = \sigma(G(\mathbf{z}, \mathbf{W})) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})) \tag{9}$$

where $\mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_D]$ (D is the number of features); ω, $\mathbf{F}_{\text{ex}}$, and δ represent the weight vector, excitation function, and ReLU function, respectively; and $\mathbf{W}_1$ and $\mathbf{W}_2$ are parameter vectors of two fully connected layers.

The ANB is utilized to adaptively normalize input features with two obtained weights ω_c^{MaSE} and ω_c^{MiSE} based on MaSE and MiSE, respectively. The maximum pooing of $\mathbf{f}_c$ performed in the MaSE is defined as

$$\mathbf{z}_c^{\max} = \max_{i,j} \mathbf{f}_c(i, j) \tag{10}$$

where $\mathbf{z}_c^{\max}$ represents the squeezed spatial information based on the global maximum pooing.

MaSE produces a channel weight ω^{MaSE} as

$$\boldsymbol{\omega}^{\text{MaSE}} = \mathbf{F}_{\text{ex}}(\mathbf{z}^{\max}, \mathbf{W}) = \sigma(g(\mathbf{z}^{\max}, \mathbf{W})) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z}^{\max})) \tag{11}$$

where $\boldsymbol{\omega}^{\text{MaSE}}$ and $\mathbf{z}^{\max}$ are $[\boldsymbol{\omega}_1^{\text{MaSE}}, \boldsymbol{\omega}_2^{\text{MaSE}}, \ldots, \boldsymbol{\omega}_D^{\text{MaSE}}]$ and $[\mathbf{z}_1^{\max}, \mathbf{z}_2^{\max}, \ldots, \mathbf{z}_D^{\max}]$, respectively.

MiSE produces a channel weight ω_c^{MiSE}

$$\boldsymbol{\omega}^{\text{MiSE}} = \mathbf{F}_{\text{ex}}(\mathbf{z}^{\min}, \mathbf{W}) = \sigma(G(\mathbf{z}^{\min}, \mathbf{W})) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z}^{\min})) \tag{12}$$

where $\boldsymbol{\omega}^{\text{MiSE}}$ and $\mathbf{z}^{\max}$ are $[\boldsymbol{\omega}_1^{\text{MaSE}}, \boldsymbol{\omega}_2^{\text{MaSE}}, \ldots, \boldsymbol{\omega}_D^{\text{MaSE}}]$ and $[\mathbf{z}_1^{\max}, \mathbf{z}_2^{\max}, \ldots, \mathbf{z}_D^{\max}]$, respectively. $\mathbf{z}_c^{\min}$ is defined as

$$\mathbf{z}_c^{\min} = \min_{i,j} \mathbf{f}_c(i, j). \tag{13}$$

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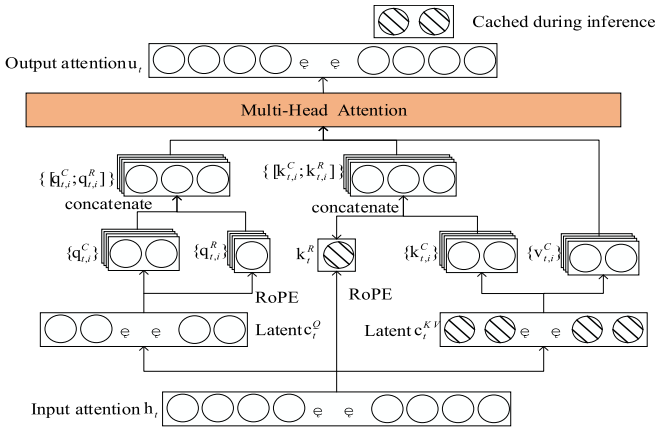


Fig. 4. MLA mechanism.

Finally, the output feature named $\mathbf{N} = [\mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_D]$ is calculated as

$$\mathbf{N} = \mathbf{f}_N(\mathbf{X}) = \frac{\mathbf{X} - \boldsymbol{\omega}^{\text{MiSE}}}{\boldsymbol{\omega}^{\text{MaSE}} - \boldsymbol{\omega}^{\text{MiSE}}}. \tag{14}$$

2) MLA Mechanism: The attention mechanism is used to extract important object features in multiple heads. Traditional attention mechanisms generate attention maps in only one head, limiting the performance of extracting features. Therefore, it is unsuitable to use traditional attention mechanisms to extract features of deep-sea fishes confused with the surroundings.

Different from channel attention mechanisms, the MLA [38] consists of multihead attention [39] and group query attention [40]. Therefore, the MLA can use each head to compute attention with its own latent vectors without applying compression to the query (Q). The structure of the MLA is depicted in Fig. 4.

The MLA combines compression of keys (K) and values (V) through low-rank compression, effectively reducing the overhead of the KV cache. The tth input attention with the embedding dimension d, attention head number n_h, and attention head dimension d_h is used to implement low-rank compression on KV

$$\mathbf{c}_t^{KV} = \mathbf{W}^{DKV}\mathbf{h}_t, \quad \mathbf{c}_t^{KV} \in R^{d_c} \tag{15}$$

$$\left[\mathbf{k}_{t,1}^C; \mathbf{k}_{t,2}^C; \ldots; \mathbf{k}_{t,n}^C\right] = \mathbf{k}_t^C = \mathbf{W}^{UK}\mathbf{c}_t^{KV}\mathbf{W}^{DKV} \in R^{d_c \times d} \tag{16}$$

$$\left[\mathbf{v}_{t,1}^C; \mathbf{v}_{t,2}^C; \ldots; \mathbf{v}_{t,n}^C\right] = \mathbf{v}_t^C = \mathbf{W}^{UV}\mathbf{c}_t^{KV} \tag{17}$$

where $\mathbf{c}_t^{KV}$ is the compressed underlying vector of K and V; d_c denotes the compressing dimension of K and V; $\mathbf{W}^{DKV}$ denotes a reduced dimension projection matrix; and $\mathbf{W}^{UK}$ and $\mathbf{W}^{UV}$ are the ascending projection matrixes of K and V, respectively.

Low-rank compression on query is conducted to reduce the active memory footprint during training. The compressed latent vector $\mathbf{c}_t^Q$ is defined as

$$\mathbf{c}_t^Q = \mathbf{W}^{DQ}\mathbf{h}_t, \quad \mathbf{c}_t^Q \in R^{d_c'}, \mathbf{W}^{DQ} \in R^{d_c' \times d} \tag{18}$$

$$\left[\mathbf{q}_{t,1}^C; \mathbf{q}_{t,2}^C; \ldots; \mathbf{q}_{t,n}^C\right] = \mathbf{q}_t^C = \mathbf{W}^{UQ}\mathbf{c}_t^Q, \quad \mathbf{W}^{DQ} \in R^{d_c' \times d} \tag{19}$$

where d_c' represents the compressing dimension of Q, $\mathbf{W}^{DQ}$ denotes the reduced projection matrix of Q, and $\mathbf{W}^{UQ}$ denotes the ascending projection matrix of Q.

Rotary position encoding (RoPE) designed based on relative position coding by absolute position coding is adopted to obtain the matrixes of Q and K, named $\mathbf{q}_t^R$ and $\mathbf{k}_t^R$, respectively,

$$\left[\mathbf{q}_{t,1}^R; \mathbf{q}_{t,2}^R; \ldots; \mathbf{q}_{t,n}^R\right] = \mathbf{q}_t^R = \text{RoPE}(\mathbf{W}^{QR}\mathbf{c}_t^Q)$$
$$\mathbf{W}^{QR} \in R^{d_h^R n_h \times d_c'} \tag{20}$$

$$\left[\mathbf{k}_{t,1}^R; \mathbf{k}_{t,2}^R; \ldots; \mathbf{k}_{t,n}^R\right] = \mathbf{k}_t^R = \text{RoPE}(\mathbf{W}^{KR}\mathbf{h}_t)$$
$$\mathbf{W}^{KR} \in R^{d_h^R \times d} \tag{21}$$

where $\mathbf{W}^{QR}$ and $\mathbf{W}^{KR}$ denote matrixes to generate decoupled Q and K, respectively, and d_h^R is the head dimension of decoupled Q and K.

The final Q matrix $\mathbf{q}_{t,i}$ and K matrix $\mathbf{k}_{t,i}$ are generated as

$$\mathbf{q}_{t,i} = \left[\mathbf{q}_{t,i}^C; \mathbf{q}_{t,i}^R\right], \quad \mathbf{k}_{t,i} = \left[\mathbf{k}_{t,i}^C; \mathbf{k}_t^R\right]. \tag{22}$$

The output of each attention head $\mathbf{o}_{t,i}$ is calculated as

$$\mathbf{o}_{t,i} = \sum_{j=1}^{t} \text{Softmax}\left(\frac{\mathbf{q}_{t,i}^{\mathrm{T}}\mathbf{k}_{t,i}}{\sqrt{d_h + d_h^R}}\right)\mathbf{v}_{j,i}^c. \tag{23}$$

The multiple attention heads are mapped to the unified space through the final linear mapping layer, obtaining the final attention $\mathbf{u}_t$

$$\mathbf{u}_t = \mathbf{W}^O[\mathbf{o}_{t,1}; \mathbf{o}_{t,2}; \ldots; \mathbf{o}_{t,n}] \tag{24}$$

where $\mathbf{W}^O$ is the output projection matrix to perform linear transformation.

Thus, the MLA can extract features in multiple attention heads with group query attention to effectively distinguish objects from surroundings.

3) ONWD Loss Function: Different from the CIOU loss function, the NWD [41] loss function comprehensively gains the feature distributions of the bounding boxes by effectively evaluating the overlap between bounding boxes. Therefore, the sensitivity of IOU to the positional deviation of tiny objects can be reduced.

It is essential to use the 2-D Gaussian distribution to model the bounding boxes due to the irregular shapes of fish targets. In the modeling processing, a bounding box $R = (cx, cy, w, h)$, where cx, cy, w, and h are the center horizontal coordinate, vertical coordinate, width, and height, respectively, is adopted. The inscribed ellipse of R is defined as

$$\frac{(x - cx)^2}{\left(\frac{w}{2}\right)^2} + \frac{(x - cx)^2}{\left(\frac{w}{2}\right)^2} = 1. \tag{25}$$

The probability density of 2-D Gaussian distribution is calculated as

$$f(X|\mu, \Sigma) = \frac{\exp\left(-\frac{1}{2}(X - \mu)^{\mathrm{T}}\Sigma^{-1}(X - \mu)\right)}{2\pi\sqrt{|\Sigma|}} \tag{26}$$

where X, μ, and Σ refer to the coordinates, mean, and covariance of distribution, respectively. When $(X - \mu)^{\mathrm{T}}\Sigma^{-1}(X - \mu)$

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equals 1, the bounding box is modeled as a 2-D Gaussian distribution $D = (\mu, \Sigma)$

$$\mu = \begin{bmatrix} cx \\ cy \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{bmatrix}. \tag{27}$$

The optimized second-order Wasserstein distance based on two bounding boxes $R_a = (cx_a, cy_a, w_a, h_a)$ and $R_b = (cx_b, cy_b, w_b, h_b)$ is designed as

$$W_2^2(D_a, D_b) = \left\| \left[cx_a, cy_a, \frac{w_a}{2}, \frac{h_a}{2} \right]^{\mathrm{T}}, \left[cx_b, cy_b, \frac{w_b}{2}, \frac{h_b}{2} \right]^{\mathrm{T}} \right\|_2^2. \tag{28}$$

The exponential function is utilized to convert $W_2^2(D_a, D_b)$ into a similarity measure ranging from 0 to 1

$$\mathrm{NWD}(D_a, D_b) = \exp\left(-\frac{\sqrt{W_2^2(D_a, D_b)}}{C} \right) \tag{29}$$

where C is a constant related to the dataset.

The NWD loss function L_{NWD} is defined as

$$L_{\mathrm{NWD}} = 1 - \mathrm{NWD}(D_a, D_b). \tag{30}$$

However, the NWD loss function fails to consider the deviations of bounding features, positions, and shapes between the prediction bounding and ground-truth bounding boxes, decreasing the detection accuracy of small objects. To solve these problems, the IOU loss function L_{IOU} and the CIOU loss function L_{CIOU} are introduced to combine with L_{NWD} to propose the ONWD loss function L_{ONWD}

$$L_{\mathrm{ONWD}} = \lambda(1 - L_{\mathrm{IOU}}) + \lambda L_{\mathrm{CIOU}} + (1 - 2\lambda)(1 - L_{\mathrm{NWD}}) \tag{31}$$

where λ is a parameter set to 0.2. The parameter is determined via grid search over two predefined arrays: [0.1, 0.3, 0.5, 0.7, 0.9] and [0.2, 0.4, 0.6, 0.8, 1.0]. The two acquired best values, 0.1 and 0.2, are identified from the arrays, and the larger one (0.2) is selected as the loss function parameter.

4) AKConv and AKC2f: The original convolution kernel in the CBS block owns a large parameter number, restricting the ability to achieve real-time and lightweight performance.

To solve the problems of the original convolution kernel, each CBS block is replaced with the AKConv [42], which incorporates a convolutional kernel featuring adjustable parameters, including the number of parameters, sampling sizes, and sampling shapes. Thus, the sampling size and shape are adaptively adjusted based on variations in target shapes and sizes, allowing for a reduction in the parameter number. The structure of AKConv is shown in Fig. 5.

In the AKConv, the channel number, height, and width of the input image are C, H, and W, respectively. First, Conv2d is used to obtain the initial sampling shape of the input image. Then, the offsets are employed to adapt the convolutional kernel shape to image features by adjusting the initial sampling shape. Finally, the feature map, resampled based on the adjusted sampling shape, undergoes reshaping, convolution, normalization, and a SiLU activation function to obtain the output image.

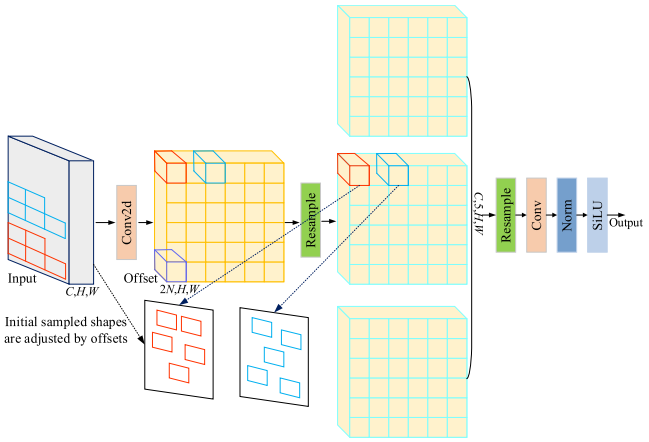


Fig. 5. Structure of AKConv.

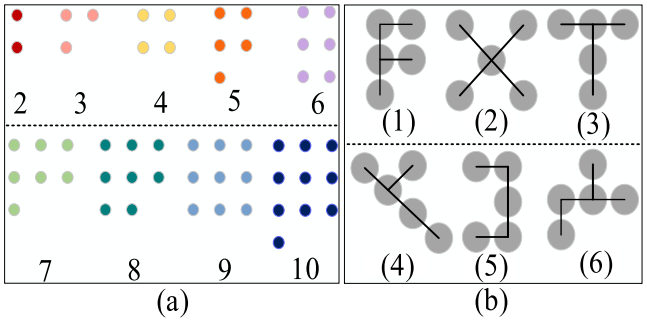


Fig. 6. Sampling process. (a) Initial sampling shapes for arbitrary sizes. (b) Initial sampling shapes for size 5 × 5.

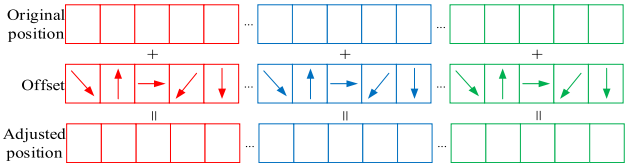


Fig. 7. Sampling shapes adjusted by offsets.

In the sampling process, unfixed sizes and shapes of the kernel are adjusted based on the density and features of targets, effectively reducing the number of parameters. AKConv automatically calculates the optimal kernel sizes to accurately match the shapes and sizes of targets, thereby enhancing feature extraction performance. Additionally, AKConv utilizes diverse image regions to efficiently extract target features using various initial sampling shapes within a 5 × 5 sampling grid. The initial sampling shapes are shown in Fig. 6. Furthermore, AKConv uses offsets to adjust the convolution kernel's position in response to local feature variations across different image regions, enabling better adaptation to non-rigid fish shapes, as depicted in Fig. 7. However, the detection accuracy may be limited owing to the structure simplification of AKConv.

To balance both the detection accuracy and speed, each C2f module in the backbone and neck is replaced with an AKC2f module, which is generated by converting the original bottleneck into the AKbottleneck. The structures of the AKC2f and C2f are shown in Fig. 8.

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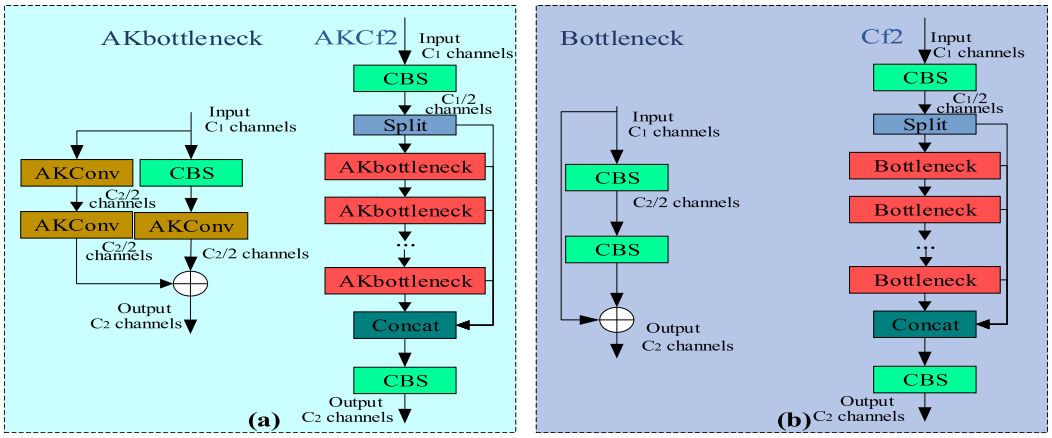


Fig. 8. Structures of (a) AKC2f and (b) C2f.

In the AKbottleneck, three original CBS blocks in one branch are replaced with two AKConv blocks, while another AKConv block and an original CBS block are introduced into the other branch. In this way, AKbottleneck can utilize two branches to perform convolution operation, enabling it to capture similarities and differences between features in different paths to extract useful target features. Meanwhile, the parameter number of the AKC2f is reduced compared with the original C2f. Therefore, the application of both AKConv and AKC2f can achieve a good balance between both detection accuracy and speed.

IV. Experiments and Results

A. Dataset Preparation

The dataset used in this article contains 1200 deep-sea fish images selected from the dataset of "labeled fishes in the wild" [43], which consists of Sebastes Spp fishes and other bottom fishes caught by underwater vehicles during the surveys of rockfish in the Southern California Bight from 2000 to 2012. In this dataset, the total number of fish targets is 3640.

The labeling software of LabelImg is used to label all the images. The labeled images are divided into the training set and validation set at a ratio of 5:1. The examples of sample labeling are shown in Fig. 9.

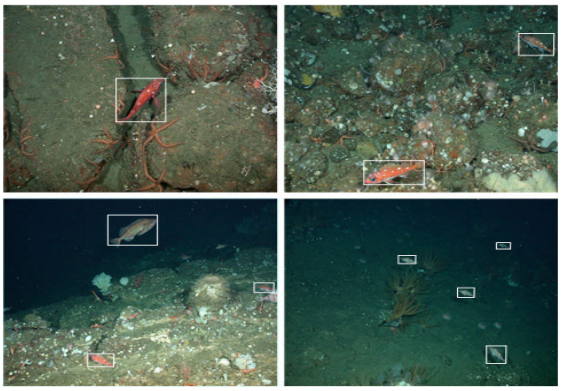


Fig. 9. Examples of sample labeling.

TABLE I
Training Environment

Training environment	Configuration
Operating system	Windows 10
CPU	Intel(R) Core(TM) i5-13400F
GPU	NVIDIA GeForce RTX 3060
Accelerated environment	CUDA 11.6.0, CUDNN 11.6
Programming language	Python 3.9.7

TABLE II
Training Parameters

Training parameters	Values
Epochs	200
Batch-size	16
Image-size	640 × 640
Initial learning rate	0.01
Optimizer	SGD

B. Experimental Environment

In this article, the model training is operated on the desktop computer to obtain the weight file, which is used to detect deep-sea fishes. The training environment and training parameters are shown in Tables I and II, respectively. The weight file and MOA-YOLO are deployed on the Nvidia Jetson Xavier NX. The parameters of Nvidia Jetson Xavier NX are shown in Table III.

C. Evaluation Metrics

The detection ability of the MOA-YOLO algorithm is evaluated in terms of the precision (P), recall (R), mean average precision (mAP), parameter number (Param), floating-point operating per second (FLOPs), and frames per second (FPS).

P and R are the estimation of the correct proportion among all samples and prediction of the correct proportion of samples among all real samples, respectively. The expressions of P and R are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{32}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{33}$$

where TP is the number of true positives, FN is the number of false negatives, and FP is the number of false positives.

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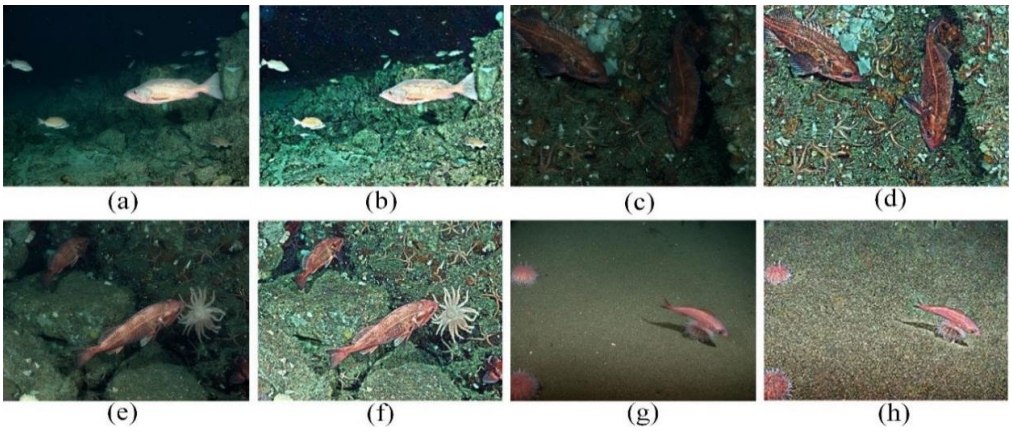


Fig. 10. Enhancement results of deep-sea targets. (a), (c), (e), and (g) Original images. (b), (d), (f), and (h) Enhanced images.

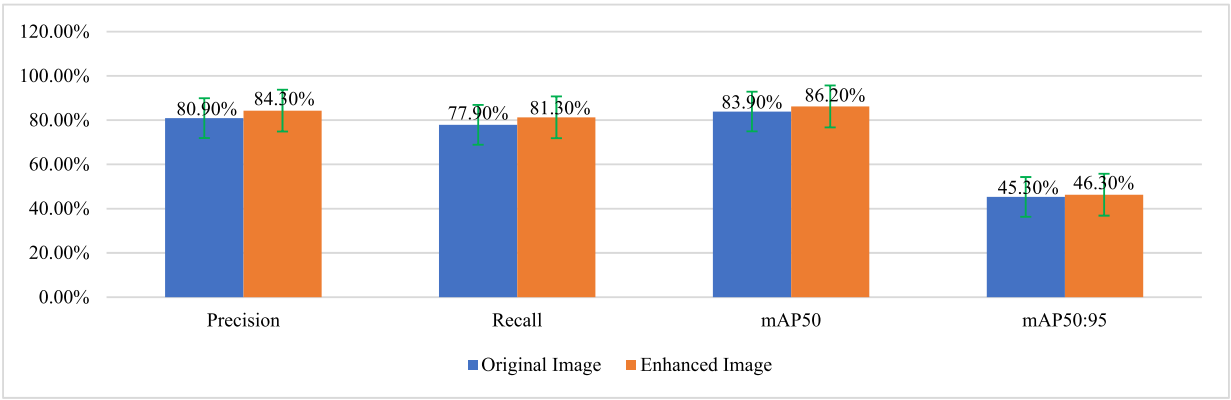


Fig. 11. Impact of image enhancement on detection accuracy.

mAP is defined as

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} \text{AP}_i \tag{34}$$

$$\text{AP} = \int_0^1 P(R)dR \tag{35}$$

where n is the number of categories and AP_i is the average precision of each category.

Additionally, Param, FLOPs, and FPS are used to measure the detection speed, floating-point calculation speed, and size of the model, respectively.

D. Effects of Image Enhancement

The underwater image enhancement algorithm based on ASNet is used to enhance different deep-sea images, as shown in Fig. 10. Compared with the original images, the color and contrast of the enhanced images are greatly corrected.

The impact of image enhancement on detection accuracy is shown in Fig. 11. The underwater image enhancement algorithm can improve the precision, recall, mAP_{50}, and $\text{mAP}_{50:95}$ of MOA-YOLO by 4.2%, 4.4%, 2.7%, and 2.2%, respectively, which proves that the designed underwater enhancement algorithm obviously improves the detection accuracy of MOA-YOLO.

E. Comparison of Different Attention Mechanisms

The MLA is introduced into the front of each detecting head to extract object features. The effectiveness of MLA is evaluated in comparison with SE [37] and efficient channel attention (ECA) [44].

As can be seen from Fig. 12, MLA improves the precision, recall, mAP_{50}, and $\text{mAP}_{50:95}$ by 7.3%, 2.4%, 3.4%, and 2.4%, respectively. In contrast, SE improves the precision by 1.0% but decreases the recall, mAP_{50}, and $\text{mAP}_{50:95}$ by 3.5%, 4.9%, and 10.4%, respectively. Moreover, ECA improves the precision by 1.9% but decreases the recall, mAP_{50}, and $\text{mAP}_{50:95}$ by 6.2%, 6.8%, and 10.6%, respectively. MLA achieves a better effect on feature extraction, which contributes to distinguishing objects from the surroundings.

To visualize the effects of attention mechanisms, the feature maps produced by the model without attention mechanisms, as well as those incorporating SE, ECA, and MLA, are compared. As shown in the red bounding boxes of Fig. 13, the feature maps generated by the MLA-equipped model highlight more fish target regions, represented by lighter shades. This visually demonstrates the superior feature extraction capability of MLA.

F. Comparison of Loss Functions

The ONWD loss function improves the ability to detect small objects. The feasibility of ONWD is verified in comparison with CIOU and NWD.

The results in Fig. 14 show that ONWD displays the best performance on all evaluation metrics and achieves the precision, recall, mAP_{50}, and $\text{mAP}_{50:95}$ of 84.3%, 81.3%, 86.2%, and 46.3%, respectively. The CIOU achieves 79.0%,

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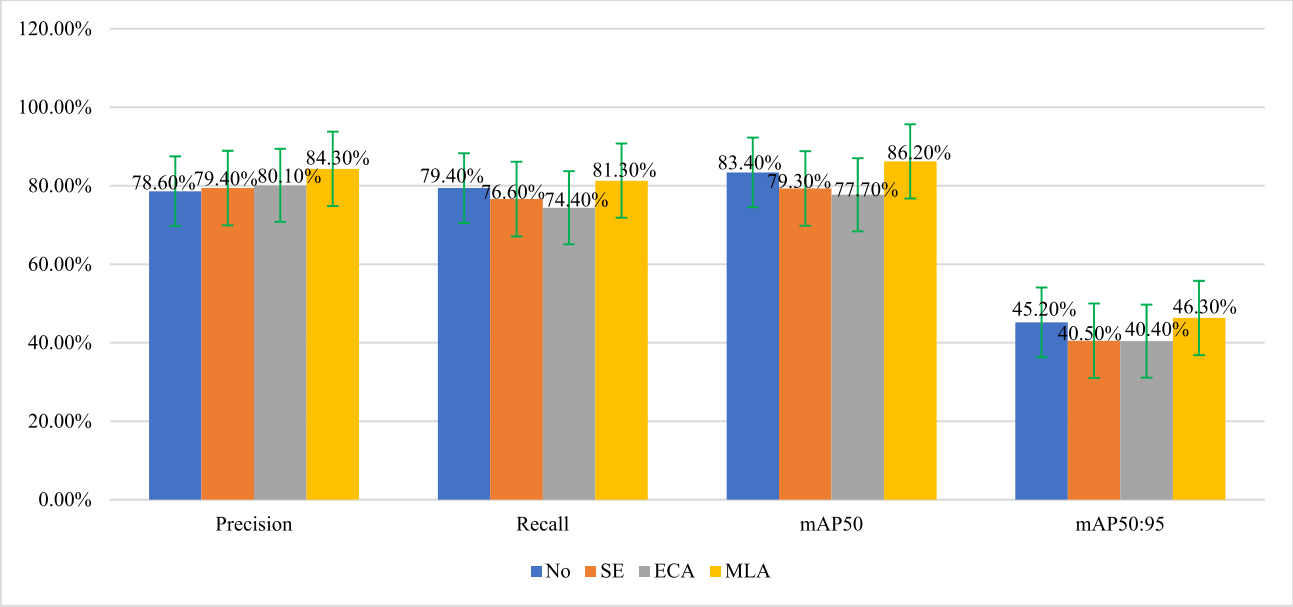


Fig. 12. Comparison of attention mechanisms.

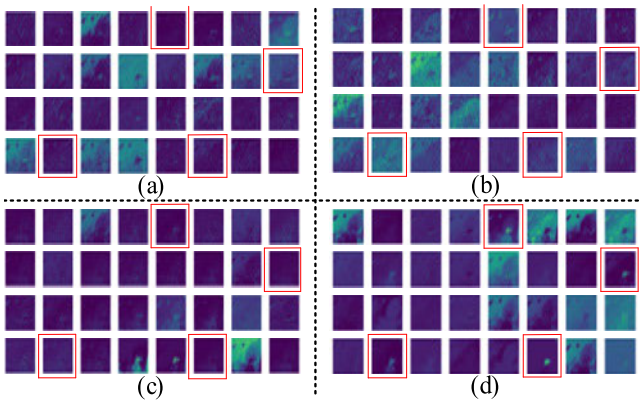


Fig. 13. Feature map visualization. (a) No. (b) SE. (c) ECA. (d) MLA.

79.4%, 84.2%, and 45.9% in the precision, recall, mAP_{50}, and $mAP_{50:95}$, respectively, and NWD achieves 83.8%, 79.3%, 85.8%, and 45.8% in the precision, recall, mAP_{50}, and $mAP_{50:95}$, respectively. It confirms that ONWD achieves better detection performance.

The evaluation loss curves of CIOU, NWD, and ONWD are shown in Fig. 15. It can be seen that the model with ONWD achieves the lowest loss values in the training process, which proves the effectiveness of ONWD in reducing image information loss.

G. Comparison of Different Utilization Methods of AKConv and AKC2f

The real-time and accurate performance of combining AKConv and AKC2f (AKConv-AKC2f) is proved in comparison with AKConv. As can be observed from Fig. 16, AKConv-AKC2f improves the precision, recall, and FPS by 3.0%, 2.8%, and 16.6%, respectively, and reduces parameters by 2.0%. However, AKConv improves the precision, recall, and FPS by only 0.7%, 1.6%, and 16.0%, respectively, and reduces parameters by only 1.9%. The results suggest that AKConv-AKC2f effectively achieves real-time performance of detection while maintaining high detection accuracy.

TABLE III
PARAMETERS OF NVIDIA JETSON XAVIER NX

Name	Nvidia Jetson Xavier NX
CPU	6-core NVIDIA Carmel ARM v8.2 64-bit CPU 6 MB L2 + 4 MB L3
GPU	384-core NVIDIA Volta GPU with 48 Tensor Cores
Memory	16 GB 128-bit LPDDR4x 59.7 GB/s
AI performance	21 TOPS
Storage	16 GB EMMC + 128 GB SSD
Size	69.6mmx45m
Power	15-20 W

TABLE IV
RESULTS OF DIFFERENT COMPRESSION DIMENSIONS

Compression dimensions	Precision	Recall
2	81.9%	79.8%
3	82.6%	80.2%
4	84.3%	81.3%
5	83.2%	80.1%
6	80.2%	78.8%

H. Comparison of Different Compression Dimensions of the MLA

The impact of varying compression dimensions of the MLA on the performance of MOA-YOLOv10 is analyzed in Table IV. The results show that a compression dimension of 4 yields the highest precision and recall, indicating minimal feature information loss. Thus, a compression dimension of 4 is the most suitable choice for the MLA.

I. Ablation Experiments

To verify the effectiveness of each improved module of MOA-YOLO, ablation experiments are conducted. The experimental results are shown in Table V and Fig. 17.

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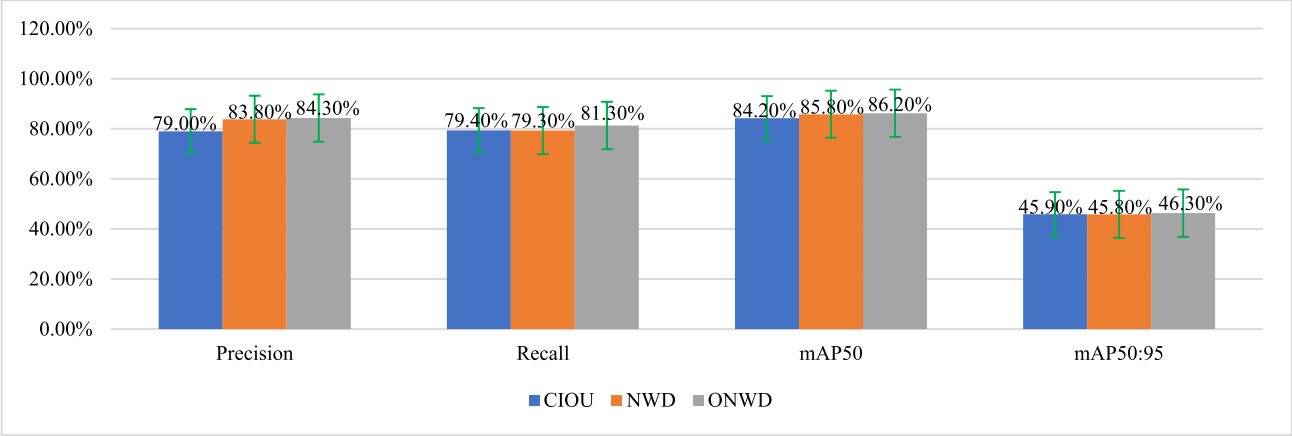


Fig. 14. Comparison of loss functions.

TABLE V
RESULTS OF ABLATION EXPERIMENTS

Models	MLA	ONWD	AKConv +AKC2f	Precision	Recall	mAP_{50}	$mAP_{50:95}$	Param (M)	FPS
Model 1				80.9%	71.5%	79.8%	43.3%	7.22	65.4
Model 2			√	80.0%	80.9%	83.4%	45.1%	7.07	79.5
Model 3		√	√	78.6%	79.4%	83.4%	45.2%	7.07	79.5
Model 4		√		81.4%	78.4%	85.1%	45.8%	7.22	65.4
Model 5	√			82.6%	81.2%	86.2%	46.5%	7.27	63.5
Model 6	√	√		81.8%	79.1%	83.9%	45.6%	7.27	63.5
Model 7	√		√	79.0%	79.4%	84.2%	45.9%	7.12	78.3
Model 8	√	√	√	84.3%	81.3%	86.2%	46.3%	7.12	78.3

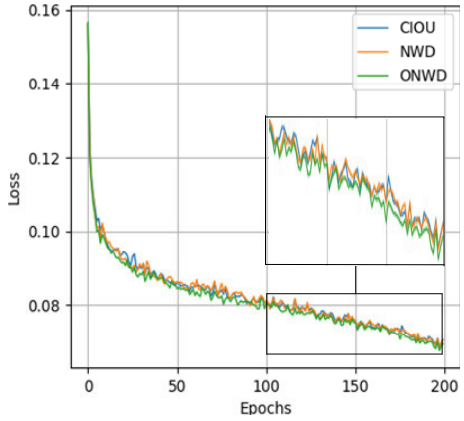


Fig. 15. Evaluation loss curves of loss functions.

Table V demonstrates the results of the ablation experiments. The addition of MLA improves the precision, recall, mAP_{50}, and $mAP_{50:95}$ by 2.1%, 13.6%, 8.0%, and 7.4%, respectively. When the CIOU loss function is replaced with ONWD, the precision, recall, mAP_{50}, and $mAP_{50:95}$ of MOA-YOLO are improved by 0.6%, 9.7%, 6.6%, and 5.3%, respectively. By introducing AKConv and AKC2f, the recall, mAP_{50}, $mAP_{50:95}$, and FPS of MOA-YOLO are improved by 13.1%, 2.1%, 4.2%, and 21.6%, respectively. When using MLA and ONWD simultaneously, the precision, recall, mAP_{50}, and $mAP_{50:95}$ of MOA-YOLO are improved by 1.1%, 10.6%, 5.1%, and 5.3%, respectively. When combining MLA, AKConv, and AKC2f, the recall, mAP_{50}, $mAP_{50:95}$, and FPS of MOA-YOLO are improved by 11.0%, 5.5%, 6.0%, and 19.7%, respectively. The combination of ONWD, AKConv, and AKC2f improves the recall, mAP_{50}, $mAP_{50:95}$, and FPS of MOA-YOLO by 11.0%, 4.5%, 4.4%, and 21.6%, respectively. In contrast, when combining MLA, ONWD, AKConv, and AKC2f, the precision, recall, mAP_{50}, $mAP_{50:95}$, and FPS of MOA-YOLO are improved by 4.2%, 13.7%, 8.0%, 6.9%, and 19.7%, respectively. Additionally, the parameter number is decreased by 1.3%.

Fig. 17 shows the performance of each model in the training process. The performance of the MOA-YOLO model is superior to those of other models, demonstrating the effectiveness of MLA, ONWD, AKConv, and AKC2f.

J. Comparison of Different Algorithms

Deep-sea fish detection is implemented on the Nvidia Jetson Xavier NX embedded in the AUV. To validate the effectiveness of MOA-YOLO deployed on the embedded platform with limited computational resources, MOA-YOLO is compared with Faster-CNN, SSD, YOLOv3, YOLOv4, YOLOv6s, YOLOv7, YOLOv8s, and YOLOv10s. The results in Table VI reveal that the MOA-YOLO algorithm achieves the highest values of precision, recall, mAP_{50}, $mAP_{50:95}$, and FPS. Compared with the other algorithms, the precision, recall, mAP_{50}, $mAP_{50:95}$, and FPS of MOA-YOLO are improved by over 3.1%, 13.7%, 8.0%, 6.9%, and 8.8%, respectively. Meanwhile, MOA-YOLO achieves less parameters and a lower FLOPs value than the

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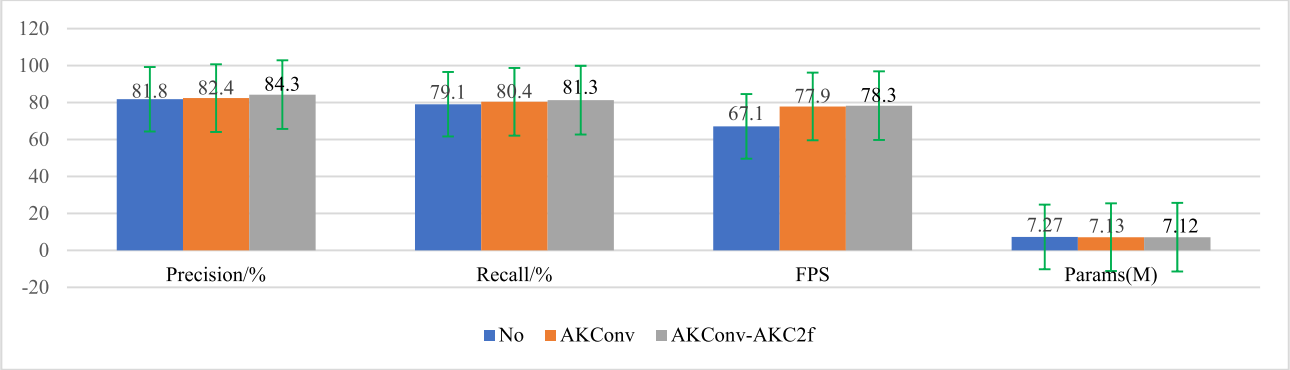


Fig. 16. Comparison of utilization methods of AKConv and AKC2f.

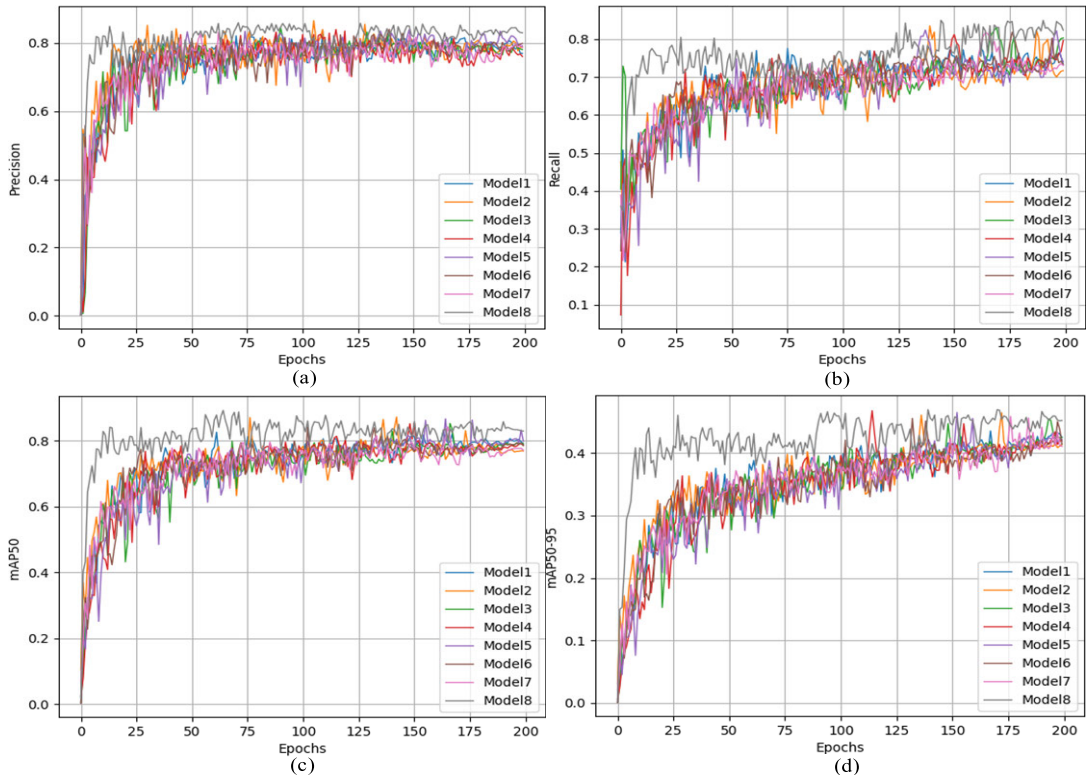


Fig. 17. Performance of each model in the training process. (a) Precision change curve. (b) Recall change curve. (c) Curve of change in mAP_{50}. (d) Curve of change in $mAP_{50:95}$.

comparative algorithms. In conclusion, MOA-YOLO surpasses mainstream algorithms in both accuracy and real-time performance by introducing MLA, ONWD, AKConv, and AKC2f.

The camera of the AUV captures deep-sea fishes in different scenes. Thus, to validate the detection performance in different scenes, the above algorithms are conducted in low-light scenes, blurred scenes, and turbid scenes.

Fig. 18 depicts the detection results in low-light scenes. Faster-CNN produces two false detections. The detection results of SDD and YOLOv3 demonstrate two missed detections. There is one missed detection in the detection results of YOLOv4, YOLOv8s, and YOLOv10s. YOLOv6s and YOLOv7 detect two fishes to be one fish. In contrast, MOA-YOLO accurately detects all the fishes in the image.

Fig. 19 shows the detection results in blurred scenes. The detection results of Faster-CNN show that fishes and surroundings are detected to be one fish. SSD detects only one fish and causes a false detection. There are many missed detections and false detection in the detection results of YOLOv3, YOLOv4, and YOLOv7. The detection results of YOLOv6s show that several fishes are detected to be one fish. YOLOv8s produces three missed detections and one false detection. There is a missed detection in the detection results of YOLOv10. In contrast, MOA-YOLO detects all the fishes and leads to no false detections.

Fig. 20 demonstrates the detection results in turbid scenes. Faster-CNN detects five fishes but causes a false detection.

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TABLE VI
EXPERIMENTAL RESULTS OF DIFFERENT ALGORITHMS

Algorithms	Precision	Recall	mAP_{50}	$mAP_{50:95}$	Param (M)	FLOPs/G	FPS
Faster-CNN	84.0%	79.2%	74.3%	37.2%	136.69	213.3	9.6
SSD	76.9%	56.4%	55.1%	30.7%	26.30	87.5	30.6
YOLOv3	75.5%	67.2%	72.7%	35.6%	61.50	154.5	22.0
YOLOv4	79.4%	69.8%	74.6%	38.5%	64.44	142.0	21.1
YOLOv6s	81.8%	62.5%	70.6%	36.0%	17.19	44.1	26.1
YOLOv7	76.9%	64.8%	69.8%	37.2%	36.49	103.2	24.9
YOLOv8s	78.9%	71.2%	76.6%	39.3%	11.12	28.4	32.7
YOLOv10s	80.9%	71.5%	79.8%	43.3%	7.22	28.4	32.8
MOA-YOLO	84.3%	81.3%	86.2%	46.3%	7.12	27.7	35.7

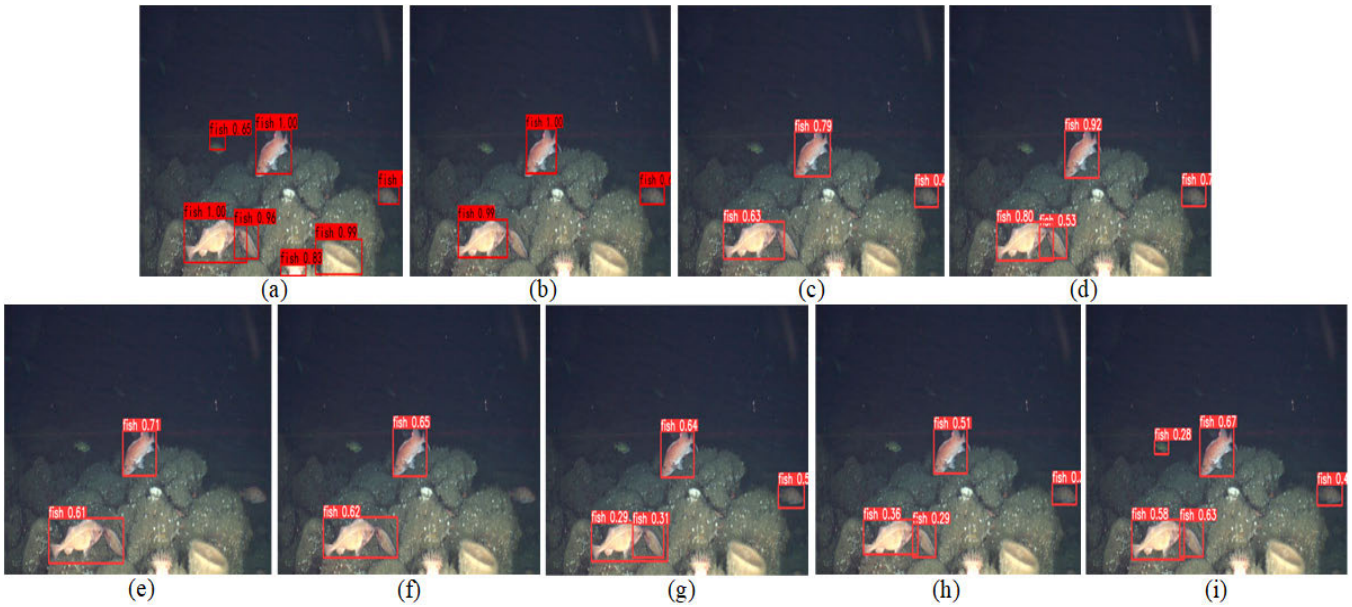


Fig. 18. Deep-sea fish detection in low-light scenes. (a) Faster-CNN. (b) SSD. (c) YOLOv3. (d) YOLOv4. (e) YOLOv6s. (f) YOLOv7. (g) YOLOv8s. (h) YOLOv10s. (i) MOA-YOLO.

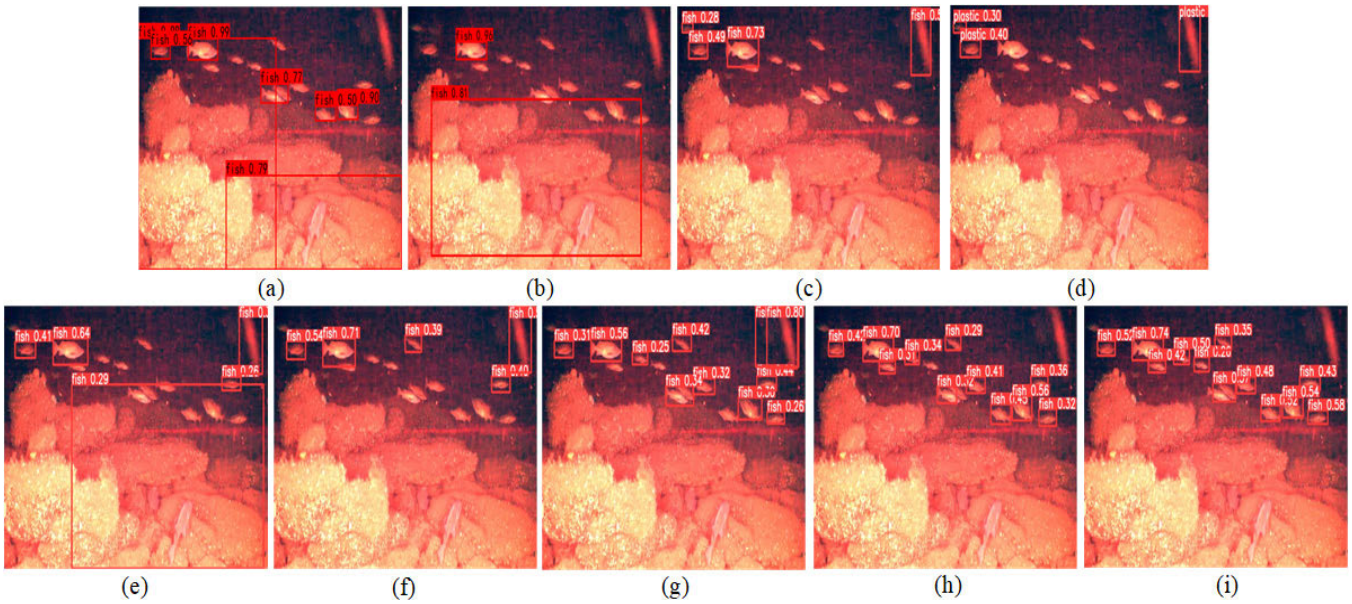


Fig. 19. Deep-sea fish detection in blurred scenes. (a) Faster-CNN. (b) SSD. (c) YOLOv3. (d) YOLOv4. (e) YOLOv6s. (f) YOLOv7. (g) YOLOv8s. (h) YOLOv10s. (i) MOA-YOLO.

SSD detects only one fish. There are three detected fishes in the results of YOLOv3. YOLOv4, YOLOv7, and YOLOv8 detect four fishes. The detection results of YOLOv6 and YOLOv10 show that two fishes are detected. On the contrary, MOA-YOLO detects five fishes and produces no false detection, achieving the best detection effect.

These detection results show that MOA-YOLO can achieve accurate detection of deep-sea fishes in different scenes.

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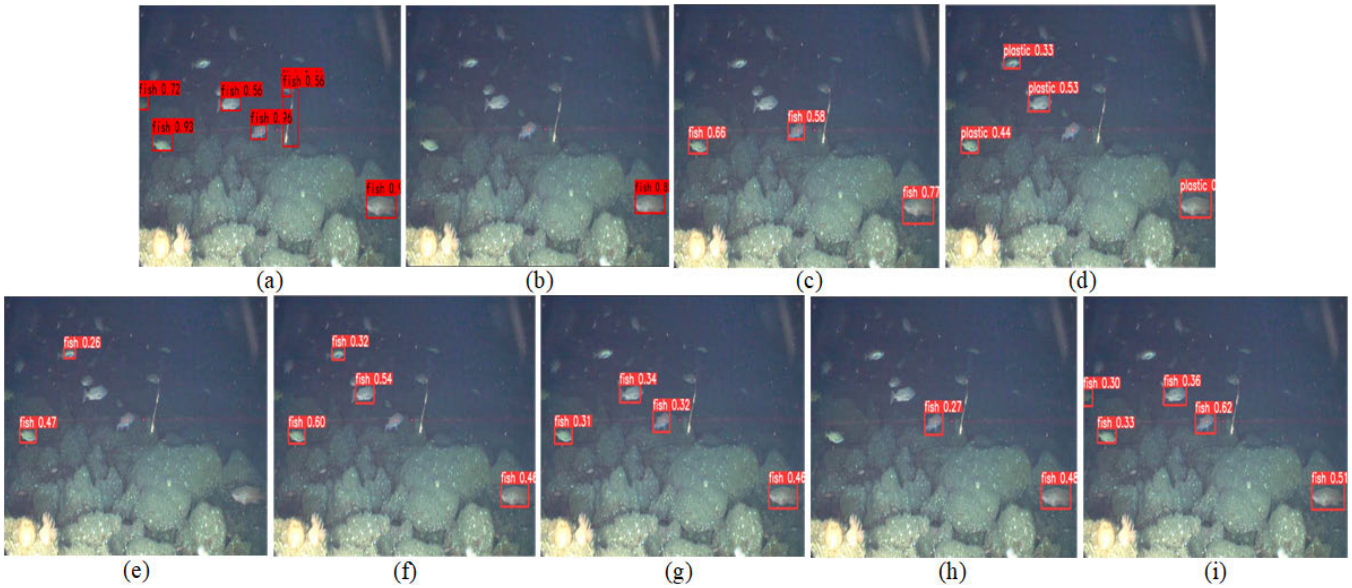


Fig. 20. Deep-sea fish detection in turbid scenes. (a) Faster-CNN. (b) SSD. (c) YOLOv3. (d) YOLOv4. (e) YOLOv6s. (f) YOLOv7. (g) YOLOv8s. (h) YOLOv10s. (i) MOA-YOLO.

TABLE VII
EXPERIMENTAL RESULTS OF SMALL TARGET DETECTION

Algorithms	Precision	Recall	AP	Param (M)	FLOPs/G	FPS
Faster-CNN	80.5%	61.1%	78.9%	136.69	213.3	9.6
SSD	80.5%	59.8%	60.1%	26.30	87.5	30.6
YOLOv3	74.2%	55.4%	63.9%	61.50	154.5	22.0
YOLOv4	76.8%	62.3%	68.9%	64.44	142.0	21.1
YOLOv6s	70.2%	67.0%	67.7%	17.19	44.1	26.1
YOLOv7	79.0%	62.5%	66.6%	36.49	103.2	24.9
YOLOv8s	78.0%	65.0%	68.8%	11.12	28.4	32.7
YOLOv10s	75.3%	65.6%	71.5%	7.22	28.4	32.8
MOA-YOLO	81.5%	68.4%	75.5%	7.12	27.7	35.7

TABLE VIII
EXPERIMENTAL RESULTS ON THE FISH4KNOWLEDGE DATASET

Algorithms	Precision	Recall	mAP_{50}	$mAP_{50:95}$	Param (M)	FLOPs/G	FPS
Faster-CNN	60.6%	94.1%	92.4%	61.2%	136.69	213.3	9.6
SSD	90.3%	87.7%	90.8%	60.8%	26.30	87.5	30.6
YOLOv3	88.1%	85.5%	89.1%	61.2%	61.50	154.5	22.0
YOLOv4	92.3%	85.5%	89.5%	60.8%	64.44	142.0	21.1
YOLOv6s	91.0%	87.4%	89.6%	60.7%	17.19	44.1	26.1
YOLOv7	92.6%	87.4%	91.1%	63.4%	36.49	103.2	24.9
YOLOv8s	91.3%	82.0%	89.9%	63.3%	11.12	28.4	32.7
YOLOv10s	90.8%	88.4%	89.9%	64.5%	7.22	28.4	32.8
MOA-YOLO	94.3%	89.3%	90.2%	66.3%	7.12	27.7	35.7

TABLE IX
EXPERIMENTAL RESULTS ON THE LIFECLEF 2015 DATASET

Algorithms	Precision	Recall	mAP_{50}	$mAP_{50:95}$	Param (M)	FLOPs/G	FPS
Faster-CNN	70.9%	88.8%	90.0%	58.5%	136.69	213.3	9.6
SSD	89.1%	86.2%	88.2%	57.6%	26.30	87.5	30.6
YOLOv3	84.8%	83.3%	86.8%	58.9%	61.50	154.5	22.0
YOLOv4	85.1%	82.6%	86.9%	58.2%	64.44	142.0	21.1
YOLOv6s	83.4%	85.6%	86.6%	57.7%	17.19	44.1	26.1
YOLOv7	84.7%	86.8%	87.5%	56.0%	36.49	103.2	24.9
YOLOv8s	85.7%	85.4%	87.7%	57.8%	11.12	28.4	32.7
YOLOv10s	82.9%	85.5%	89.1%	58.1%	7.22	28.4	32.8
MOA-YOLO	89.8%	89.1%	90.0%	60.0%	7.12	27.7	35.7

K. Extended Experiments

The performance of small target detection plays a crucial role in deep-sea fish detection. Thus, fish targets with pixel areas below 32×32 are classified as small targets to evaluate the performance of different algorithms. The proportion of small targets in the dataset is 15%. Table VII shows that MOA-YOLO surpasses mainstream algorithms in metric evaluation, where AP is an independent detection indicator for small

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TABLE X
EXPERIMENTAL RESULTS ON THE NVIDIA JETSON XAVIER TX2

Algorithms	Precision	Recall	mAP_{50}	$mAP_{50:95}$	Param (M)	FLOPs/G	FPS
Faster-CNN	84.0%	79.2%	74.3%	37.2%	136.69	213.3	8.5
SSD	76.9%	56.4%	55.1%	30.7%	26.30	87.5	28.6
YOLOv3	75.5%	67.2%	72.7%	35.6%	61.50	154.5	20.2
YOLOv4	79.4%	69.8%	74.6%	38.5%	64.44	142.0	19.0
YOLOv6s	76.9%	64.8%	69.8%	37.2%	17.19	44.1	24.2
YOLOv7	81.8%	62.5%	70.6%	36.0%	36.49	103.2	24.9
YOLOv8s	78.9%	71.2%	76.6%	39.3%	11.12	28.4	30.6
YOLOv10s	75.2%	65.5%	71.3%	35.5%	7.22	28.4	31.5
MOA-YOLO	83.8%	81.4%	84.2%	46.0%	7.12	27.7	34.1

objects. Thus, MOA-YOLO succeeds in detecting small targets like deep-sea fishes.

To verify the versatility of MOA-YOLO on different datasets, these algorithms are conducted on the Fish4Knowledge [45] and LifeCLEF 2015 [46] datasets, as shown in Tables VIII and IX, respectively. It can be seen from Tables VIII and IX that MOA-YOLO achieves the best results in almost all metrics, proving the great versatility of MOA-YOLO on different datasets.

The comprehensive results in Table X demonstrate that MOA-YOLO surpasses the other algorithms in all metrics, validating the versatility of MOA-YOLO on different platforms.

V. CONCLUSION

Deep-sea fishes are dynamic and small objects that are easily confused with the surroundings, making it hard to detect fishes in degraded deep-sea environments. Moreover, the limited computing resources of the AUV restrict the parameter number of the detection algorithm. To solve these problems, an MOA-YOLO algorithm is proposed in this article. First, an underwater image enhancement algorithm based on ASNet is adopted to correct the color and contrast of deep-sea images. Then, the MLA is added to the front of each detection head to distinguish targets from surroundings. Next, an ONWD loss function is proposed to replace the CIOU loss function to detect small objects. Finally, AKConv and AKC2f modules are introduced into the backbone and neck to achieve a lightweight model and real-time performance. Conducting experiments on the Nvidia Jetson Xavier NX, the results show that MOA-YOLO is superior to mainstream detection algorithms in deep-sea fish detection. Compared with the other algorithms, the precision, recall, mAP_{50}, $mAP_{50:95}$, and FPS of MOA-YOLO are improved by at least 3.1%, 13.7%, 8.0%, 6.9%, and 8.8%, respectively. Meanwhile, MOA-YOLO displays a lower parameter number and FLOPs value. As a result, MOA-YOLO produces accuracy, real-time, and lightweight performance for deep-sea fish detection.

Although MOA-YOLO achieves great performance in deep-sea fish detection, this study still has some shortcomings: First, MOA-YOLO produces relatively low confidence values in target detection, potentially leading to missed or uncertain predictions. Second, this study evaluates MOA-YOLO for a single fish species, without assessing its performance in detecting diverse fish types. In the future, this study will elevate confidence scores in target detection by refining the loss function of MOA-YOLO and evaluating the ability of MOA-YOLO to detect multiple fish types.

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